

Abby Browne

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EXPERIENCE:

United Launch Alliance

Centennial, CO

Systems Engineer, Vulcan Development

April 2024 to present

- Led cross-functional technical integration across multiple teams to deliver seven formal gate reviews to senior leadership and key customers, ensuring design requirements and functionality met program expectations.
- Led project management for ground systems testing, maintaining a readiness tracker, coordinating inputs across engineering leads, aligning the team on test objectives, organizing test dates, and presenting readiness status to government customers.
- Coordinated 20+ Technical Design Reviews, ensuring clarity of design intent, completeness of content, and alignment across stakeholders.
- Built and managed deliverable timelines to support a fast-paced development program, prioritizing tasks for the design team while considering five conflicting program schedules.
- Authored 100+ verification closure statements by gathering technical evidence from drawings, tests, vendor certifications, and design reviews, translating findings into documentation that verified requirements were met.
- Served as project lead and subject matter expert for a 10-person team, providing guidance and direction to meet tight deadlines for design verification efforts.

Mechanical Design Engineer, Vulcan Transportation and Handling

September 2022 to September 2024

- Developed Ground Support Equipment for rocket assembly, transportation, and maintenance through a process of conceptualization, design, analysis, technical design reviews, drawing creation and release, procurement, testing, and install.
- Created 3D models using NX and Creo to visualize early concepts, refine form and function, and improve manufacturability while maintaining design intent.
- Collaborated with cross-disciplinary teams to understand interfaces and use cases, producing design solutions that minimized cost and labor while maximizing efficiency and safety.
- Participated in trade studies to evaluate multiple design paths, balancing technical constraints, cost, and user needs.
- Generated and participated in failure analyses for hardware to identify failure modes, effects, and mitigations.

CleanRobotics Inc.

Longmont, CO

Mechanical Engineer

September 2021 to September 2022

- Led an aesthetic and UX redesign effort of TrashBot by conducting market research, sketching concepts to enhance visual appeal and usability, and developing 3D Solidworks renderings from hand-drawn concepts.
- Utilized rapid prototyping (3D printing) to iterate form, fit, and function, enabling quick evaluation of design alternatives.
- Assembled 14 full TrashBot units under tight deadlines, using various manufacturing methods while documenting processes.
- Authored user manuals in Confluence, collaborating with Customer Success to ensure clarity for customer use.
- Performed testing and maintenance on mechanical and electronic components to ensure fully functional units.

Robotics/Mechanical Engineering Intern

June 2021 to September 2021

- Improved product designs by identifying flaws, researching and demonstrating solutions with proof of concept prototypes, conducting iterative and formal design reviews, and implementing final design improvements to the product.
- Developed CAD models for new TrashBot components, including a spring-loaded belt tensioner and a 3D printed assembly jig that reduced manufacturing time.

EDUCATION:

University of Colorado Boulder, College of Engineering and Applied Science

Graduated May 2022

Bachelor of Science Mechanical Engineering

Boulder, CO

- GPA 3.83 | Dean's List Spring 2022
- President Horace M. Hale Award, Engineering Scholarship Fund, and SWE Scholarship Fund awarded for exceptional merit

RELEVANT SKILLS:

- Design & UX: Concept sketching, user-centered design, usability considerations, prototyping, design reviews, documentation.
- Software: 3D CAD Modeling (NX, Creo, Solidworks), Teamcenter, Microsoft Office, Confluence, Jira, Windchill, MATLAB, C++, Arduino.
- Prototyping & Fabrication: 3D printing, laser cutting, machining, soldering, hands-on assembly, testing.
- Engineering: Systems thinking, requirements analysis, verification & validation, failure analysis, trade studies, cross-functional collaboration.

INTERESTS:

Graphic, aesthetic, and UX design; trend research; art; travel; long-distance running; hiking; skiing; camping; knitting; baking.